

This project examines varying factors influencing paddy (rice) production and explores techniques for accurate growth predictions. Since rice is a critical agricultural commodity worldwide, improving efficiency has direct economic and food security implications. The research utilizes a dataset of environmental conditions, agricultural management practices, and production outcomes, demonstrating that combined effects of weather and agricultural practices are primary drivers of accurate forecasts. For agricultural stakeholders and policy makers, this provides a roadmap to maximize yield outcomes and stabilize the agricultural economy. To further understand the implications of the farming practices and climatic variables on yield growth, it is essential to consider previous research shows that yield is strongly connected to the interconnected roles of environmental conditions and management practices. Scholars such as Muthukumaran et al. (2023) emphasize the critical role of temperature, rainfall, humidity, and wind alongside fertilizer application, soil type, and seed rate, as these elements directly influence essential physiological processes like photosynthesis. Beyond local levels, research by Jeong et al. (2016), Sah et al. (2024), and Sahoo et al. (2024) demonstrate that these influences operate across regional and broader scales, reinforcing the importance of diverse growing conditions.

The study investigates how climate variables influence yield across growth stages, hypothesizing that these significantly affect yield outcomes, with mid-stage weather (days 31 and 90) acting as a strong predictor. Building on this, the research determines whether feature selection improves accuracy; the study hypothesizes that focusing on key variables creates more dependable tools for agricultural planning. Regarding human intervention, the study asks how management measures influence yield, hypothesizing that these practices play a key role in productivity. To address technical implementation, the research asks which machine learning methods achieve the highest accuracy, hypothesizing that approaches capturing complex

interactions, such as XGBoost, will provide more accurate estimates than traditional formulas.

Finally, the study examines the practical applications, hypothesizing that translating findings into simple productivity categories will effectively identify at-risk fields and empower decision-makers to optimize resources efficiently.

The methodology compares simplified and full feature sets and evaluates different predictive approaches to identify the most accurate model for yield estimation. Farming practices were grouped into a management intensity measure, and yield outcomes were further classified into low-, medium-, and high-yield categories to support decision-making. The results revealed that weather variables alone are weak predictors of yield, while agricultural management practices have a stronger influence on production outcomes. Models that account for the complex relationships between variables and the paddy yield produce more accurate predictions than simpler approaches. This allows fields to be grouped into performance categories that can support targeted agricultural decisions such as resource allocation and monitoring.

However, certain constraints remain; the use of aggregated climate data that may not capture local field conditions and reduced interpretability of more complex models. The data is limited to historical aspects collected rather than acknowledging external factors like sudden changes in fertilizer prices or labor shortages. Models with moderate predictive performance cannot be used for exact predictions but for broad decision-support signals. Future work should focus on integrating daily, plot-level sensor data and real-time economic indicators to refine these forecasts and improve overall model interpretability. By translating these complex machine learning results into simple productivity categories, this research provides a practical roadmap for agricultural decision-making.

References

- Muthukumaran, S., Peter, K. J., Dilipkumar, E., Savithri, S., & Senbagam, K. (2023). A hybrid machine learning model with combined wrapper feature selection techniques to improve the yield of paddy. *International Journal of Electronics and Communication Engineering*. <https://www.internationaljournalssrg.org/IJECE/paper-details?Id=512>
- Jeong, J. H., Resop, J. P., Mueller, N. D., et al. (2016). Random forests for global and regional crop yield predictions. *PLOS ONE*, 11(6). <https://doi.org/10.1371/journal.pone.0156571>
- Sah, S., Haldar, D., Singh, R., et al. (2024). Rice yield prediction through integration of biophysical parameters with SAR and optical remote sensing data using machine learning models. *Scientific Reports*, 14. <https://doi.org/10.1038/s41598-024-72624-4>
- Sahoo, R. K., Ray, S. S., & Manjunath, K. R. (2024). Advanced prediction of rice yield gaps under climate uncertainty using machine learning techniques in Eastern India. *Journal of Agriculture and Food Research*, 8, Article 100617. <https://doi.org/10.1016/j.jafr.2024.100617>
- UCI Machine Learning Repository. (2024). Paddy crop dataset. University of California, Irvine. <https://archive.ics.uci.edu/>